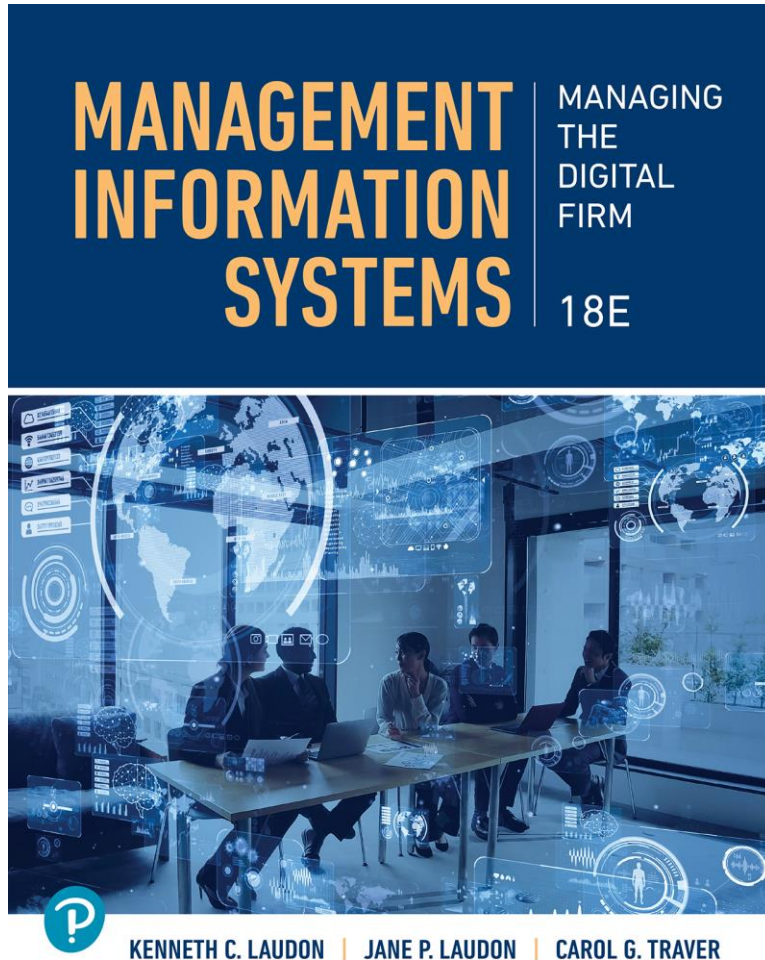


Management Information Systems: Managing the Digital Firm

Eighteenth Edition



Chapter 3

Information Systems, Organizations, and Strategy

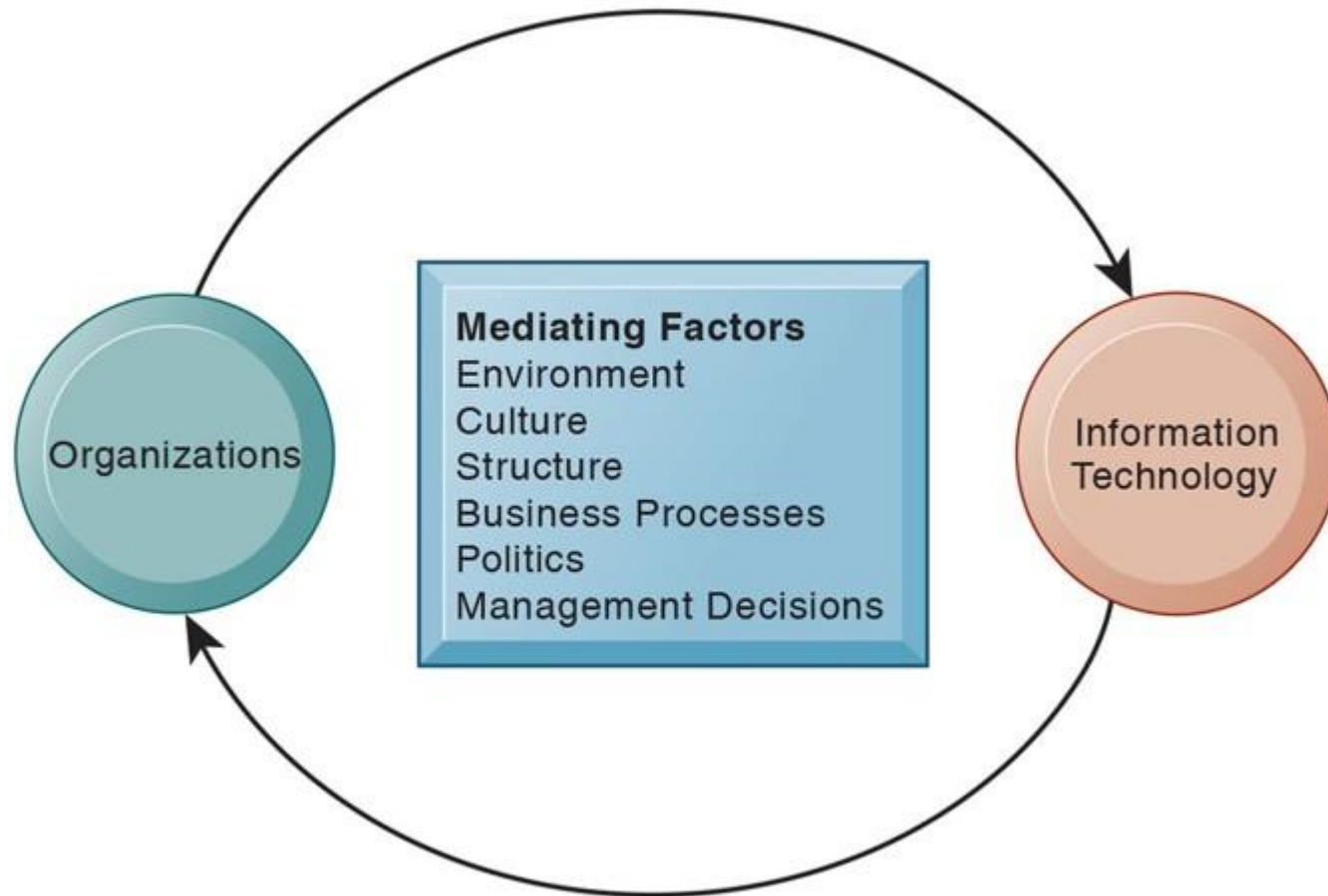
Learning Objectives

- 3.1** Discuss technical and behavioral views of organizations.
- 3.2** Discuss features of organizations that affect information systems.
- 3.3** Describe the impact of information systems on organizations.
- 3.4** Discuss Porter's competitive forces model.
- 3.5** Discuss the value chain model.
- 3.6** Discuss synergies, core competencies, and network economics.
- 3.7** Understand how AI can be used in competitive strategy.
- 3.8** Describe the challenges posed by strategic information systems.
- 3.9** Understand how the information in this chapter can help your career.

Technical and Behavioral Views of Organizations (1 of 2)

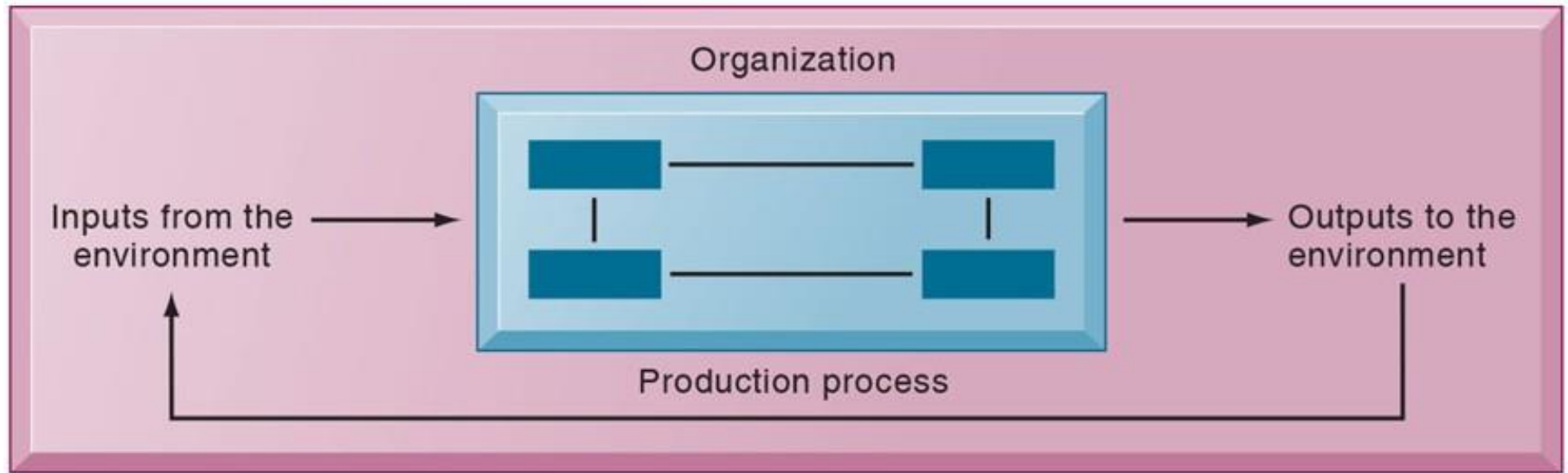
- What is an organization?
 - A stable, formal social structure that processes resources from environment to produce outputs
 - A formal legal entity with internal rules and procedures, as well as a social structure
- Behavioral definition
 - A collection of rights, privileges, obligations, and responsibilities delicately balanced over a period of time through conflict and conflict resolution

Figure 3.1 The Two-Way Relationship Between Organizations and Information Technology



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Figure 3.2 The Technical Microeconomic Definition of the Organization

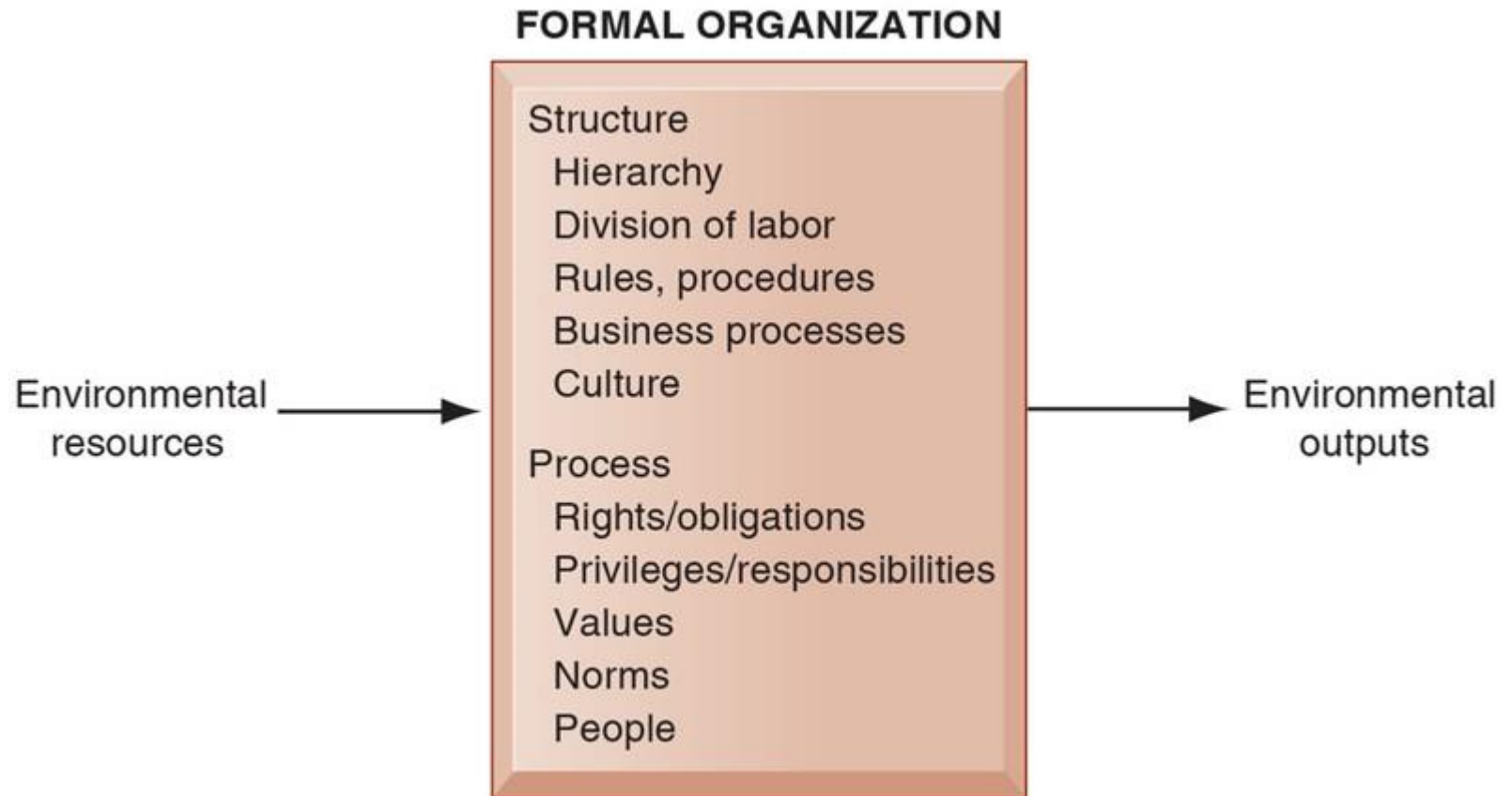


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Technical and Behavioral Views of Organizations (2 of 2)

- Organizations are formal legal entities with internal rules and procedures
 - Must abide by laws
 - Social structures
 - A collection of social elements

Figure 3.3 The Behavioral View of Organizations

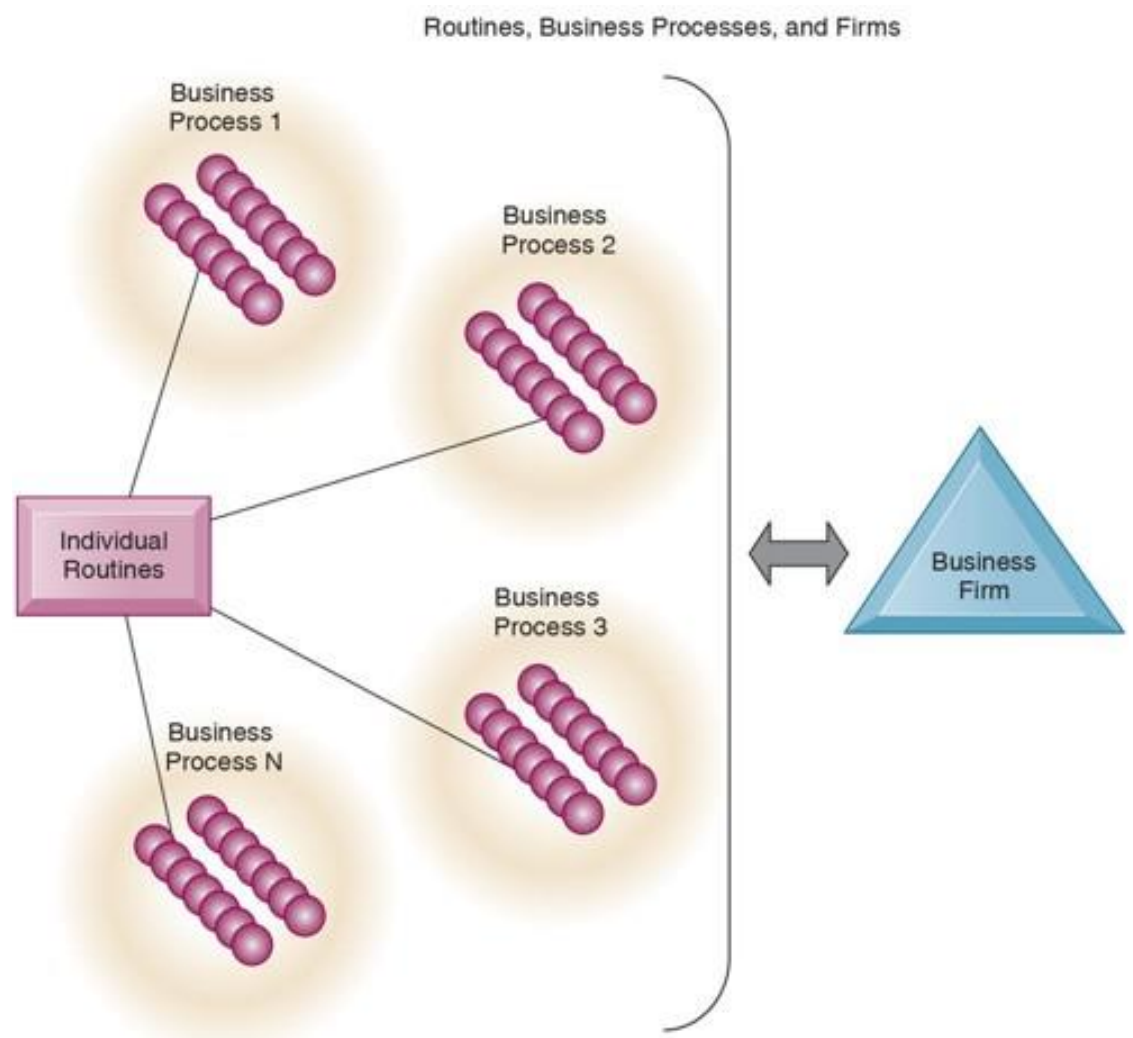


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Features of Organizations that Affect Information Systems (1 of 5)

- Routines (standard operating procedures)
 - Precise rules, procedures, and practices developed to cope with virtually all expected situations
- Business processes
 - Collections of routines

Figure 3.4 Routines, Business Processes, and Firms



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Features of Organizations that Affect Information Systems (2 of 5)

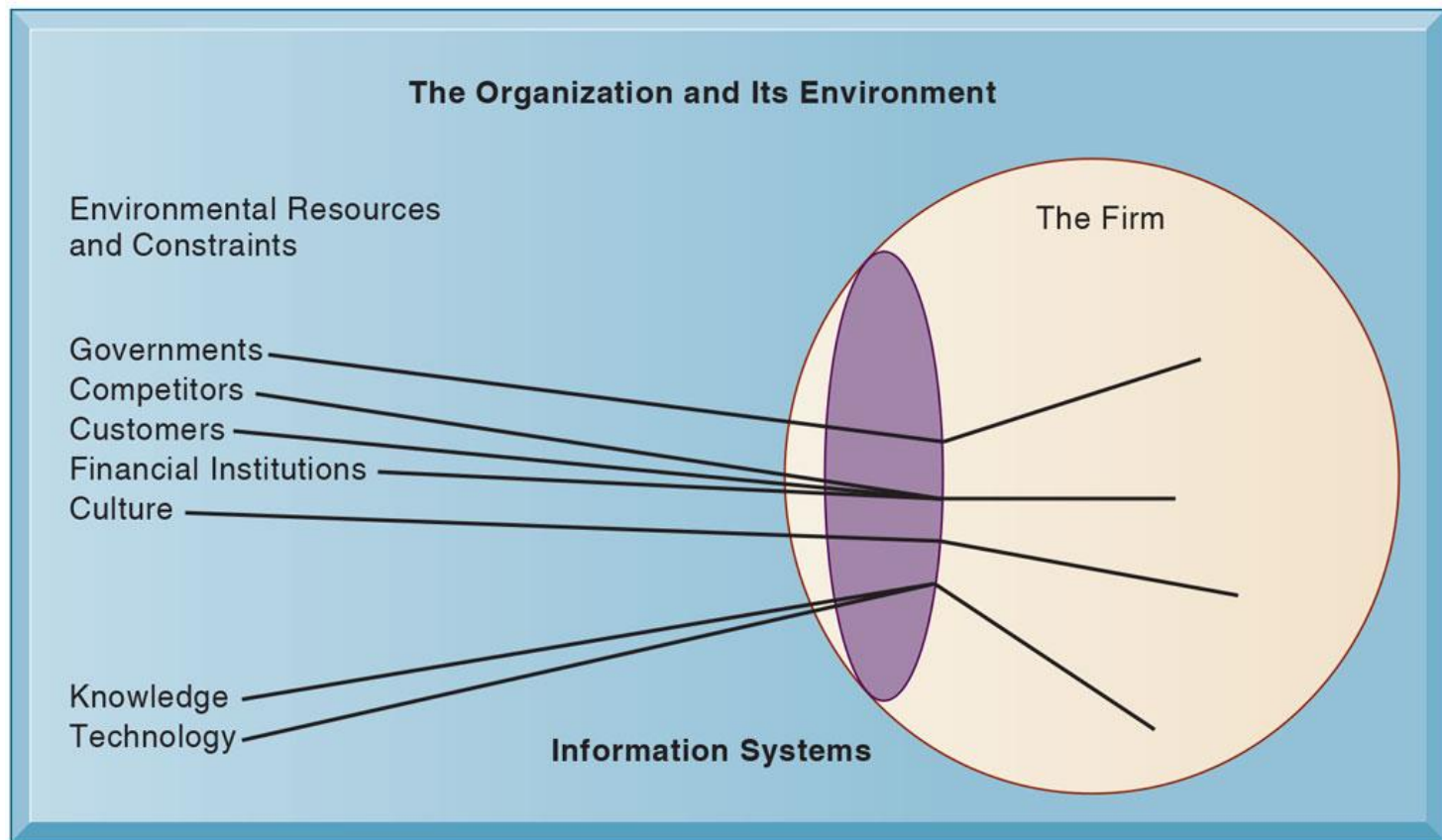
- Organizational politics
 - People in organizations, from different positions with different specialties, concerns, and perspectives, naturally have divergent viewpoints
 - The result can be political struggles for resources, competition, and conflict

Features of Organizations that Affect Information Systems (3 of 5)

- Organizational culture
 - A set of assumptions about what products the organization should produce
 - How, where, and for whom it should produce them
 - **Business processes**—the actual way business firms produce value—are usually ensconced in the organization's culture

Figure 3.4 Routines, Business Processes, and Firms

- Organizational environments
 - Environments and organizations have a reciprocal relationship



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Features of Organizations that Affect Information Systems (4 of 5)

- **Disruptive technologies** are those that substitute for and significantly reduce the market share of existing products, services, and business models
- A technology and resulting business innovation may come along and radically change the business landscape and environment

Table 3.1**Disruptive Technologies: Winners and Losers**

Technology	Description	Winners and Losers
Microprocessor chips (1971)	Thousands and eventually millions of transistors on a silicon chip	Microprocessor firms win (Intel, Texas Instruments), while transistor firms (GE) decline.
Personal computers (1975)	Small, inexpensive, but fully functional desktop computers	PC manufacturers (HP, Apple, IBM) and chip manufacturers prosper (Intel), while mainframe (IBM) and minicomputer (DEC) firms lose.
World Wide Web (1989)	A global database of digital files and "pages" instantly available	Owners of online content and news benefit, while traditional publishers (newspapers, magazines, and broadcast television) lose.
Internet music, video, TV services (1998)	Repositories of downloadable music, video, and TV broadcasts on the web	Owners of Internet platforms, telecommunications providers owning Internet backbone (ATT, Verizon), and local Internet service providers win, while content owners and physical retailers (Tower Records, Blockbuster) lose.
Software as web service	Using the Internet to provide remote access to online software	Online software services companies (Salesforce.com) win, while traditional "boxed" software companies (Microsoft, SAP, Oracle) lose or adapt.
Artificial intelligence	Using machine learning and generative AI to make decision making more rapid and accurate and to create new knowledge and content	Organizations providing AI technology and services such as NVIDIA and organizations with AI expertise win, while providers and users of less advanced technologies, along with book, film, music, and other content creators and providers, lose.

Features of Organizations that Affect Information Systems (5 of 5)

- Organizational structure
 - The structure or shape of an organization impacts the information systems in a business
 - Nature of problems with these systems often reflects the type of organizational structure

Table 3.2**Organizational Structures**

Organizational Type	Description	Examples
Entrepreneurial structure	Young, small firm in a fast-changing environment. It has a simple structure and is managed by an entrepreneur serving as its single chief executive officer.	Small start-up business
Machine bureaucracy	Large bureaucracy existing in a slowly changing environment, producing standard products. It is dominated by a centralized management team and centralized decision making.	Midsize manufacturing firm
Divisionalized bureaucracy	Combination of multiple machine bureaucracies, each producing a different product or service, all topped by one central headquarters.	<i>Fortune</i> 500 firms, such as General Motors
Professional bureaucracy	Knowledge-based organization where goods and services depend on the expertise and knowledge of professionals. Dominated by department heads with weak centralized authority.	Law firms, school systems, hospitals
Adhocracy	Task force organization that must respond to rapidly changing environments. Consists of large groups of specialists organized into short-lived multidisciplinary teams and has weak central management.	Consulting firms, such as the RAND Corporation

The Impact of Information Systems on Organizations (1 of 6)

- Over the past decade, information systems have fundamentally altered the economics of organizations
 - Has greatly increased the possibilities for organizing work
 - Theories and concepts from economics and sociology help us understand the changes brought about by IT

The Impact of Information Systems on Organizations (2 of 6)

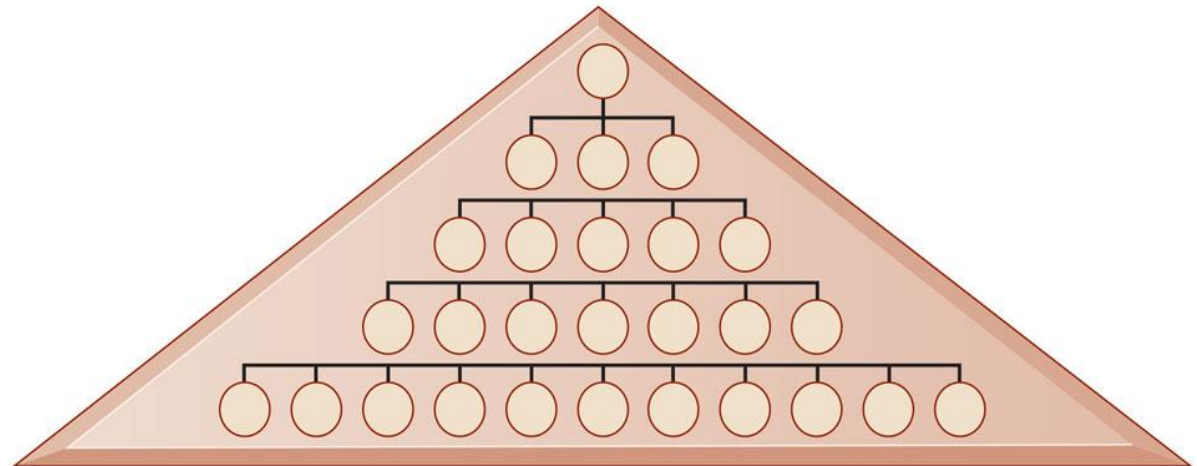
- Economic impacts
 - IT changes relative costs of capital and the costs of information
 - Information systems technology is a factor of production, like capital and labor
 - Affects the cost and quality of information
 - Changes economics of information

The Impact of Information Systems on Organizations (3 of 6)

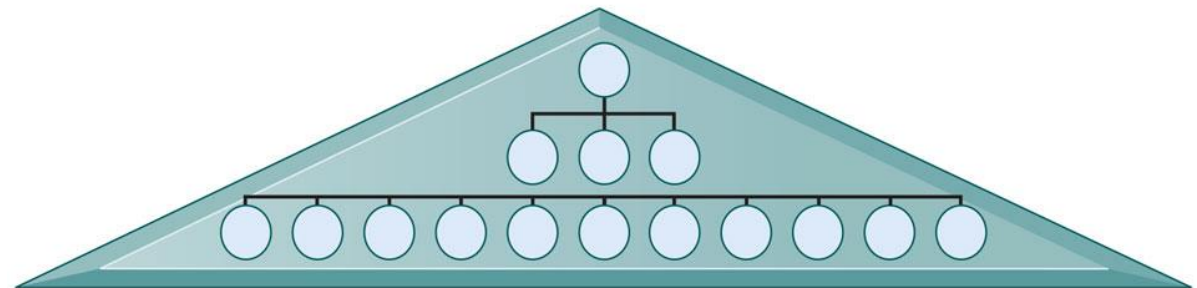
- Economic impacts
 - Information technology helps firms contract in size because it can reduce transaction costs (cost of participating in markets)
 - **Transaction costs**
 - The costs incurred when a firm buys on the marketplace what it cannot make itself
 - **Transaction cost theory**
 - Companies seek to economize on transaction costs, much as they do on production costs
 - **Agency theory**
 - The firm is viewed as a “nexus of contracts” among self-interested individuals rather than as a unified, profit-maximizing entity

Figure 3.6 Flattening Organizations

- Organizational and behavioral impacts
 - IT flattens organizations



A traditional hierarchical organization with many levels of management



An organization that has been "flattened" by removing layers of management

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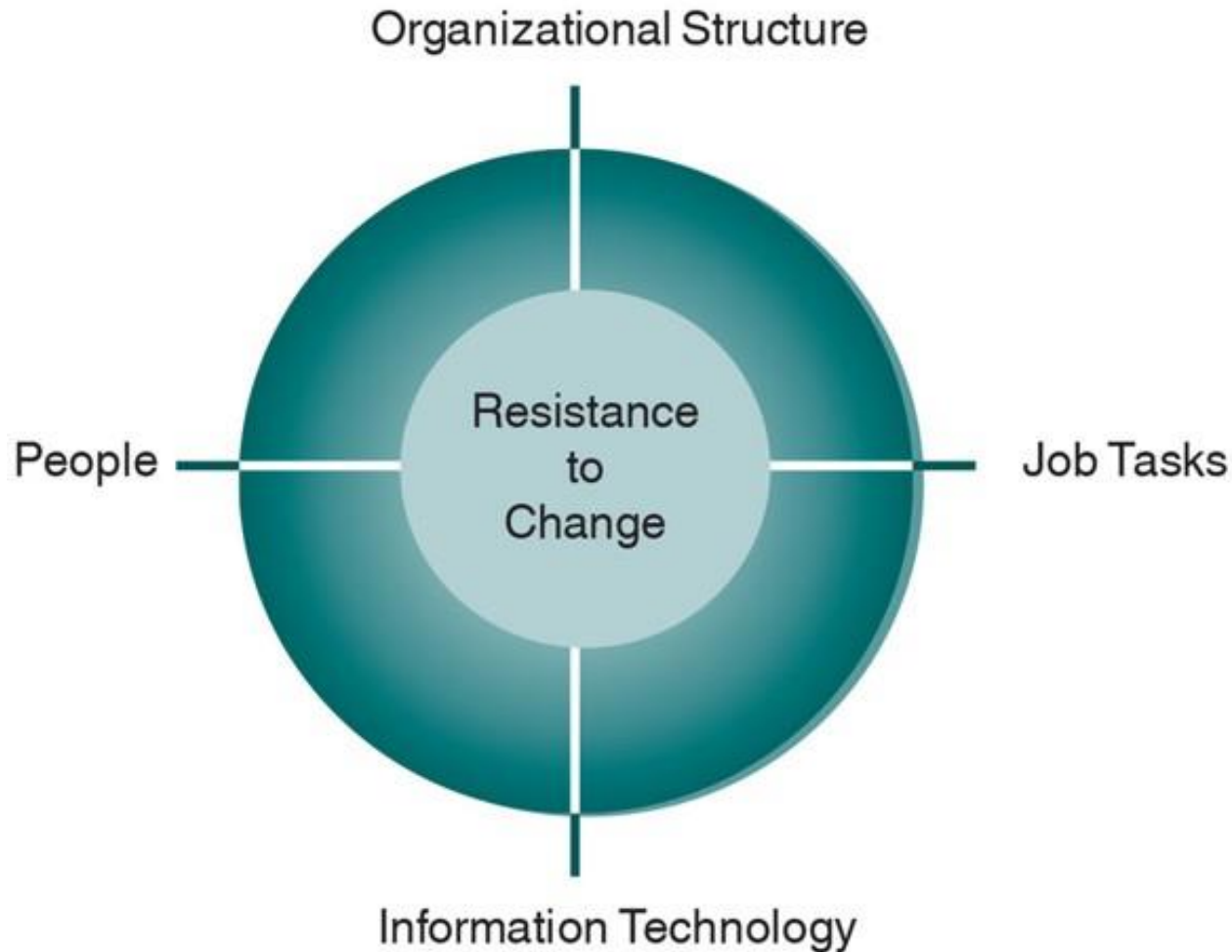
The Impact of Information Systems on Organizations (4 of 6)

- Organizational and behavioral impacts
 - Information systems affect who does what to whom, when, where, and how in an organization
 - Information systems potentially change an organization's structure, culture, business processes, and strategy, there is often considerable resistance to them when they are introduced



Supermarkets and large retail stores such as Walmart use sales data captured at the checkout counter to determine which items have sold and need to be reordered. Walmart's continuous replenishment system transmits orders to restock directly to its suppliers. The system enables Walmart to keep costs low while fine-tuning its merchandise to meet customer demands.

Figure 3.7 Organizational Resistance to Information System Innovations



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The Impact of Information Systems on Organizations (5 of 6)

Central organizational factors to consider when planning a new system include:

- The environment in which the organization must function
- The structure of the organization
 - hierarchy, specialization, routines, and business processes
- The organization's culture and politics

The Impact of Information Systems on Organizations (6 of 6)

Central organizational factors to consider when planning a new system include:

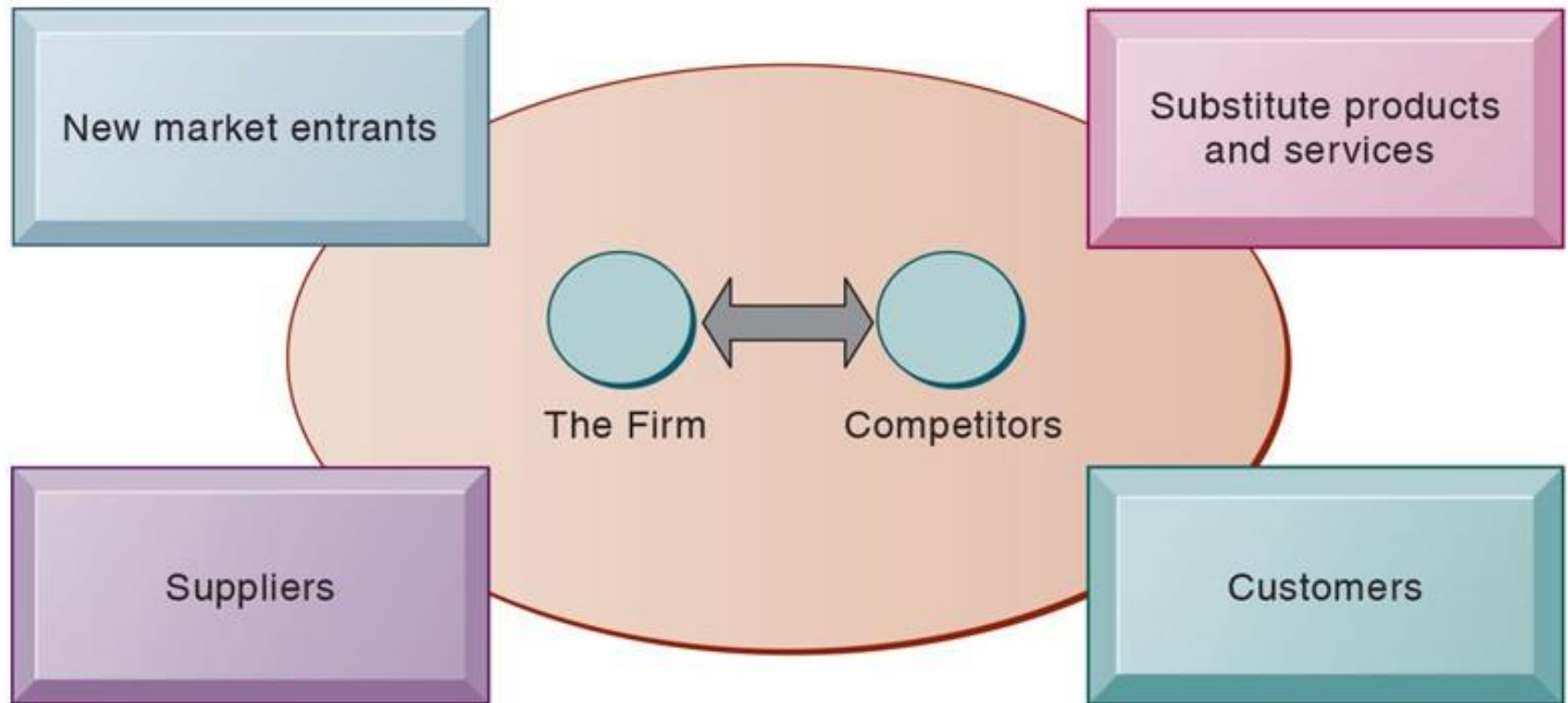
- The type of organization and its style of leadership
- The principal interest groups affected by the system and the attitudes of workers who will be using the system
- The kinds of tasks, decisions, and business processes that the information system is designed to assist

Porter's Competitive Forces Model

(1 of 3)

- Why do some firms become leaders in their industry?
- **Michael Porter's competitive forces model**
 - Provides general view of firm, its competitors, and environment
- Five competitive forces shape fate of firm:
 - Traditional competitors
 - New market entrants
 - Substitute products and services
 - Customers
 - Suppliers

Figure 3.8 Porter's Competitive Forces Model



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Porter's Competitive Forces Model

(2 of 3)

- What is a firm to do when it is faced with these competitive forces?
- Four generic strategies for dealing with competitive forces, enabled by using IT
 - Low-cost leadership
 - Product differentiation
 - Focus on market niche
 - Strengthen customer and supplier intimacy

Table 3.4**Four Basic Competitive Strategies**

Strategy	Description	Example
Low-cost leadership	Use information systems to produce products and services at a lower price than competitors while enhancing quality and level of service	Walmart
Product differentiation	Use information systems to differentiate products, and enable new services and products	Uber, Nike, Apple
Focus on market niche	Use information systems to enable a focused strategy on a single market niche; specialize	Hilton Hotels, Harrah's
Customer and supplier intimacy	Use information systems to develop strong ties and loyalty with customers and suppliers	Toyota Corporation, Amazon

Porter's Competitive Forces Model

(3 of 3)

- Transformation or threat to some industries
 - Examples: travel agency, printed encyclopedia, media
- Competitive forces still at work, but rivalry more intense
- Universal standards allow new rivals, entrants to market
- New opportunities for building brands and loyal customer bases

Table 3.5

Impact of the Internet on Competitive Forces and Industry Structure

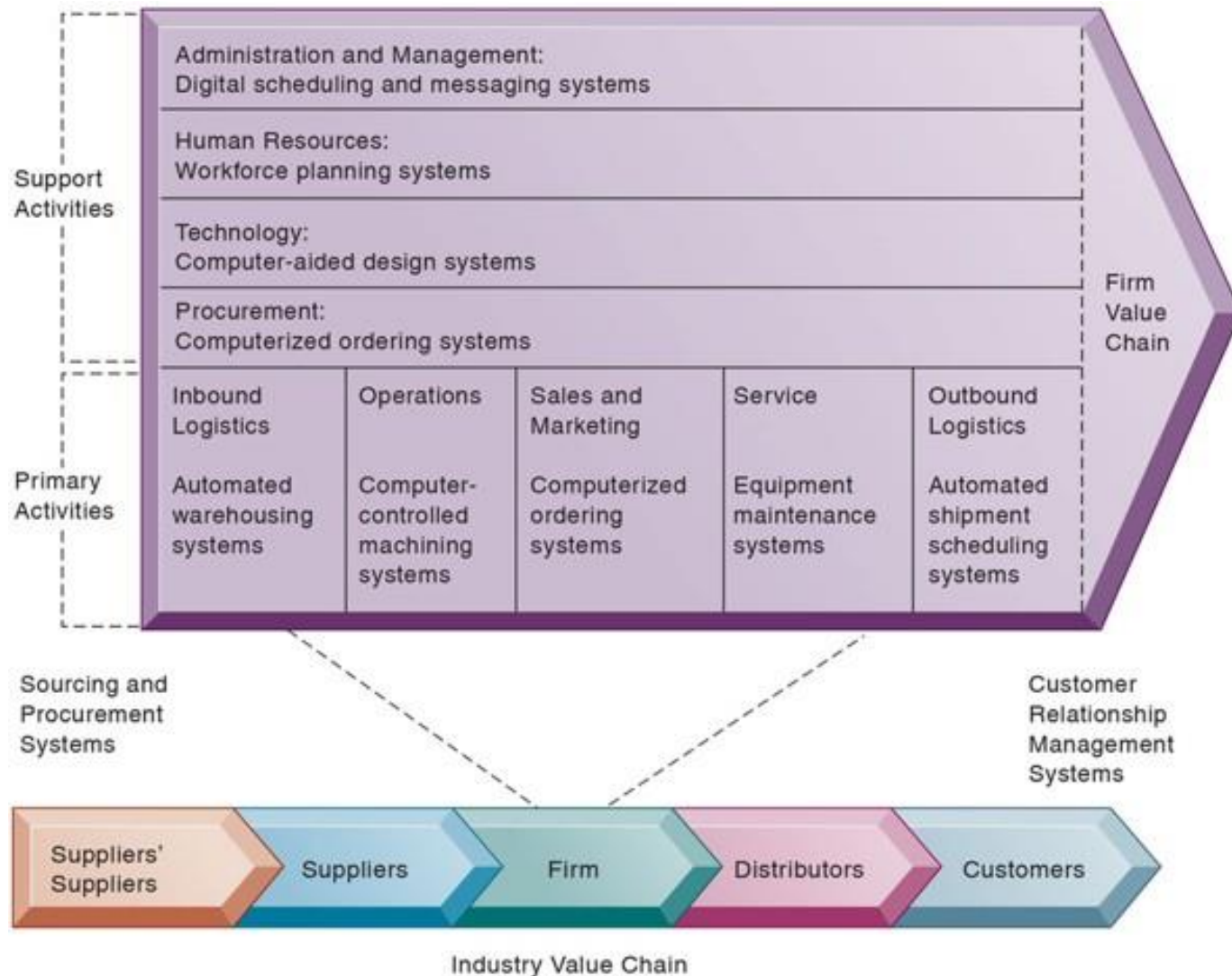
Competitive Force	Impact of the Internet
Substitute products or services	Enables substitutes to emerge with new approaches to meeting needs and performing functions
Customers' bargaining power	Availability of global price and product information shifts bargaining power to customers
Suppliers' bargaining power	Procurement over the Internet tends to raise bargaining power over suppliers; suppliers can also benefit from reduced barriers to entry and from the elimination of distributors and other intermediaries standing between them and their users
Threat of new entrants	Reduces barriers to entry, such as the need for a sales force, access to channels, and physical assets; provides a technology for driving business processes that makes other things easier to do
Positioning and rivalry among existing competitors	Widens the geographic market, increasing the number of competitors and reducing differences among competitors; makes it more difficult to sustain operational advantages; puts pressure to compete on price

The Value Chain Model

(1 of 4)

- The value chain model highlights specific activities in the business
 - Where competitive strategies can best be applied
 - Where information systems are most likely to have a strategic impact

Figure 3.9 The Value Chain Model



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The Value Chain Model

(2 of 4)

- Primary activities

- Most directly related to the production and distribution of the firm's products and services, which **create value for the customer**

- Support activities

- Make the delivery of the primary activities possible
- Consist of activities related to the organizational infrastructure
 - Administration and management
 - Human resources
 - Technology
 - Procurement

The Value Chain Model

(3 of 4)

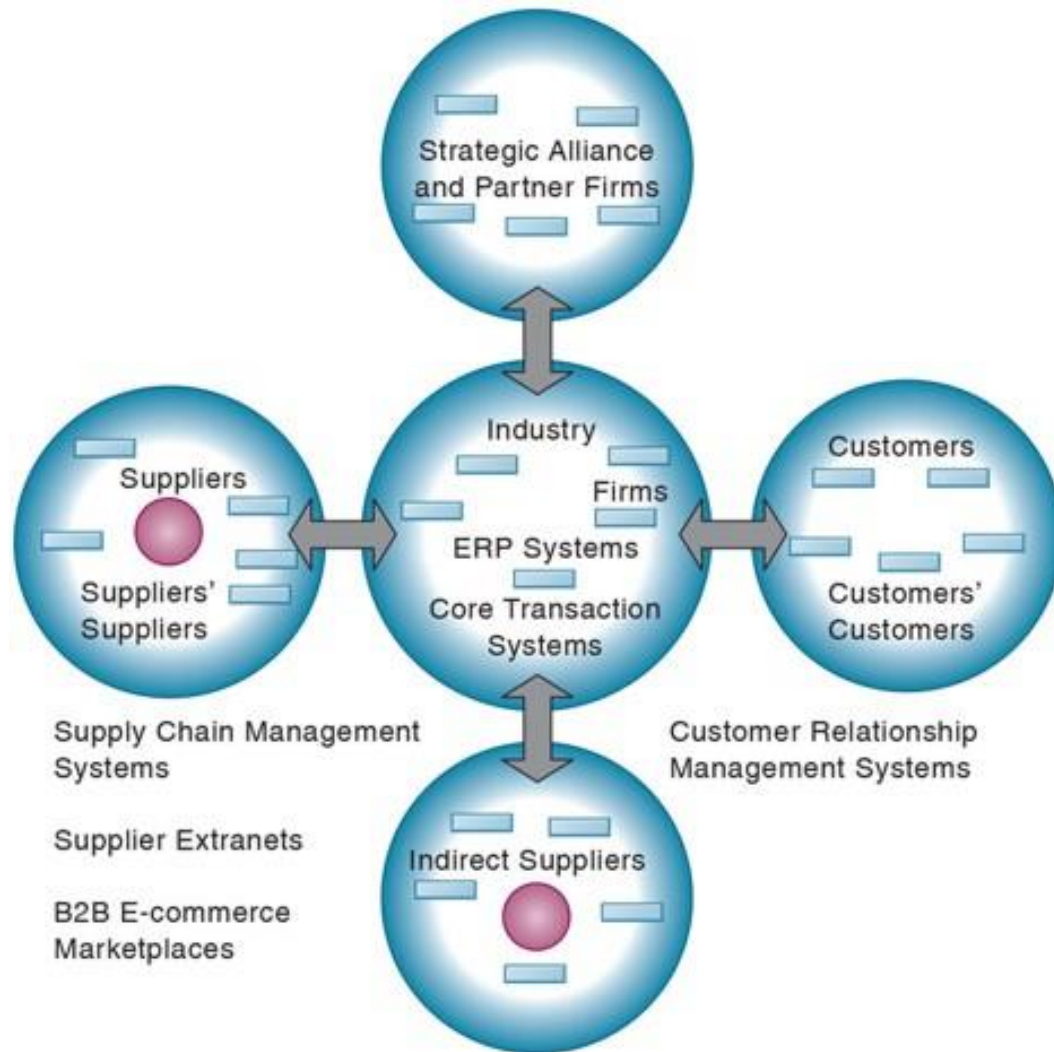
- Benchmarking
 - Involves comparing the efficiency and effectiveness of a business' processes to strict standards
 - Then measuring performance against those standards
- Best practices
 - Usually identified by consulting companies, research organizations, government agencies, and industry associations
 - The most successful solutions or problem-solving methods for consistently and effectively achieving a business objective

The Value Chain Model

(4 of 4)

- Firm's value chain is linked to value chains of suppliers, distributors, customers
- **Industry value chain**
- **Value web**
 - Collection of independent firms using highly synchronized IT to coordinate value chains to produce product or service collectively
 - More customer driven, less linear operation than traditional value chain

Figure 3.10 The Value Web



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Synergies

- When output of some units are used as inputs to others, or organizations pool markets and expertise
- Examples
 - Merger of Bank of New York and JP Morgan Chase
 - Bank of America and Countrywide Financial Corporation

Core Competencies

- Activity for which firm is an industry leader
- Relies on knowledge, experience, and sharing this across business units
- Example: Procter & Gamble's intranet and directory of subject matter experts

Network-Based Strategies (1 of 2)

- Take advantage of firm's abilities to network with one another
- Include use of:
 - Network economics
 - Virtual company model
 - Business ecosystems

Network-Based Strategies (2 of 2)

- Marginal cost of adding new participant almost zero, with much greater marginal gain
- Value of community grows with size
- Value of software grows as installed customer base grows
- Compare to traditional economics and law of diminishing returns

Virtual Company Model

- Virtual company
 - Uses networks to ally with other companies
 - Creates and distributes products without being limited by traditional organizational boundaries or physical locations
 - Example: Li & Fung
 - Manages production, shipment of garments for major fashion companies
 - Outsources all work to thousands of suppliers

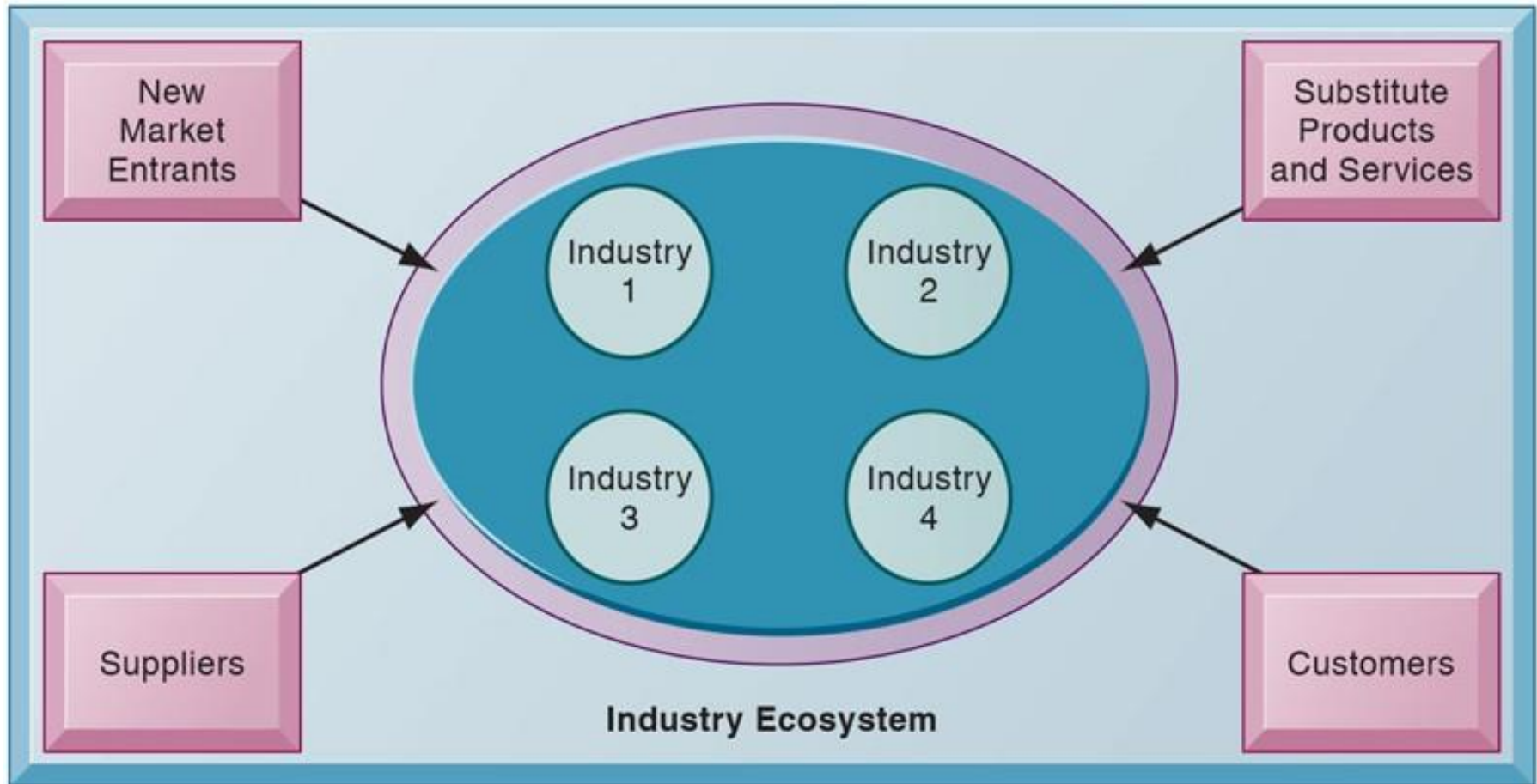
Virtual Company Model Example: Li & Fung

Fashion companies such as GUESS, Ann Taylor, Levi Strauss, and Reebok enlist Hong Kong-based Li & Fung to manage production and shipment of their garments. Li & Fung handles product development, raw material sourcing, production planning, quality assurance, and shipping. Li & Fung does not own any fabric, factories, or machines, outsourcing all of its work to a network of more than 15,000 suppliers in 40 countries all over the world. Customers place orders with Li & Fung over its private extranet. Li & Fung then sends instructions to appropriate raw material suppliers and factories where the clothing is produced. The Li & Fung extranet tracks the entire production process for each order. Working as a virtual company keeps Li & Fung flexible and adaptable so that it can design and produce the products ordered by its clients in short order to keep pace with rapidly changing fashion trends.

Business Ecosystems and Platforms

- Industry sets of firms providing related services and products
- Platforms
 - Microsoft, Facebook
- Keystone firms
- Niche firms
- Individual firms can consider how IT will help them become profitable niche players in larger ecosystems

Figure 3.11 An Ecosystem Strategic Model



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How AI Can Be Used in Competitive Strategy (1 of 2)

- AI is transforming the way information systems operate and deliver value to organizations
- AI is making it easier for companies to improve performance based on their existing business models
 - But it is also spawning new business models
 - Reshaping the competitive landscape

How AI Can Be Used in Competitive Strategy (2 of 2)

- AI technology has impacted the various competitive forces identified in Porter's model, but not entirely in the same way as the Internet
 - Substituting products or services
 - Customers' bargaining power
 - Suppliers' bargaining power
 - Threat of new entrants
 - Positioning and rivalry among existing competitors

Table 3.6**Examples of AI Applications in the Value Chain**

Value Chain Activity	AI Application
Inbound Logistics	AI in automated warehousing can maintain real-time, error-free inventory records, identify defects, and assign incoming shipments from suppliers to optimal warehouse locations.
Operations	AI identifies defects in large manufacturing runs, increasing quality while lowering costs.
Sales and Marketing	AI analyzes customer behavior to personalize marketing campaigns.
Service	AI chatbots provide quick answers to common queries from customers, sales reps, and service technicians.
Outbound Logistics	AI optimizes outbound shipment schedules, delivery routes, and route planning, reducing transportation costs.
Administration and Management	Predictive analytics can help managers anticipate demand fluctuations, optimize inventory levels, and allocate resources effectively.
Human Resources	AI algorithms predict which sourcing channels would most likely attract the right candidates for a specific job, reducing firm spending on identifying and recruiting new employees.
Technology	Firms can use AI technology to develop new products, such as noise cancelling technology for consumer products, hearing aids, earphones.
Procurement	AI can improve supplier selection by analyzing historical data, supplier performance, and market trends.

The Challenges Posed by Strategic Information Systems

- Sustaining competitive advantage
 - Competitors can retaliate and copy strategic systems
 - Systems may become tools for survival
- Aligning IT with business objectives
 - Performing strategic systems analysis
 - Structure of industry
 - Firm value chains

Management Checklist: Performing a Strategic Systems Analysis

- To identify the types of systems that provide a strategic advantage to their firms, **managers should ask the following questions:**
 - What is the structure of the industry in which the firm is located?
 - What are the business and industry value chains for this particular firm?
 - Have we aligned IT with our business strategy and goals?
Have we correctly articulated our business strategy and goals?

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